UNIT 8 · TOPIC 1(B)

Non-Ferrous and Composite Materials

Syllabus topic: Ferrous, non-ferrous, alloy and composite materials: composition, production techniques and selection for farm equipment

Non-ferrous metals

Metal / alloy	Properties	Farm-machine uses
ALUMINIUM & alloys	Light (density ~2.7 g/cm³, ⅓ of steel), corrosion-resistant, good thermal/electrical conductor; softer and costlier	Engine pistons, cylinder heads, radiators, some housings, lightweight structures
COPPER & alloys	Excellent conductor	Electrical windings, radiator tubes
BRASS (Cu–Zn)	—	Fittings, valves, radiator cores
BRONZE (Cu–Sn; also Al/phosphor bronze)	Good wear resistance + low friction	Bushes, bearings, worm wheels
ZINC	Corrosion protection	Galvanising; die-castings
LEAD	—	Battery plates, solder

Non-ferrous metals resist rust and are chosen where weight, conductivity or corrosion resistance outweigh cost.

Composite materials

A COMPOSITE combines a **MATRIX** (continuous phase) with a **REINFORCEMENT** (fibres/particles) to get properties neither has alone — typically a high strength-to-weight ratio.

Composite	Make-up	Uses
FIBRE-REINFORCED POLYMERS (FRP)	Glass (GFRP) or carbon fibres in a resin matrix — light, corrosion-proof, mouldable	Sprayer/seed/fertiliser tanks, guards, hoods, cabs, fan blades, body panels; increasingly tractor bonnets and battery housings
METAL-MATRIX composites (MMC); cermets/hard-facing	WC (tungsten-carbide) particles in a metal matrix	Extreme wear resistance on tool edges
Natural composites	Rubber/elastomer, wood	Tyres, belts, handles

- **Advantages:** light, corrosion-free, design freedom.
- **Limitations:** cost, repair and recycling.

Practice MCQs

Q16. Aluminium's density is about:

(a) 2.7 g/cm³ (⅓ of steel)
(b) 7.8 g/cm³
(c) 11 g/cm³
(d) 19 g/cm³

Q17. Aluminium alloys are used in tractors for:

(a) Plough shares
(b) Radiators, pistons, cylinder heads
(c) Crankshafts
(d) Springs

Q18. Brass is an alloy of:

(a) Cu + Sn
(b) Cu + Zn
(c) Fe + C
(d) Al + Mg

Q19. Bronze is an alloy of:

(a) Cu + Zn
(b) Cu + Sn
(c) Fe + Ni
(d) Al + Si

Q20. Bronze is preferred for bushes and bearings because of its:

(a) Magnetism
(b) Good wear resistance and low friction
(c) Light weight
(d) Transparency

Q21. Galvanising for corrosion protection uses a coating of:

(a) Zinc
(b) Lead
(c) Tin
(d) Nickel

Q22. Copper is chosen for windings and radiator tubes because it is a good:

(a) Insulator
(b) Electrical/thermal conductor
(c) Abrasive
(d) Magnet

Q23. A composite material consists of:

(a) One pure metal
(b) A matrix plus a reinforcement
(c) Two gases
(d) Only polymer

Q24. FRP (fibre-reinforced polymer) is used in farm machinery for:

(a) Crankshaft
(b) Sprayer/seed tanks, guards, hoods
(c) Gears
(d) Bearings

Q25. The chief advantage of composites is:

(a) Highest density
(b) High strength-to-weight and corrosion resistance
(c) Low cost always
(d) Easy recycling

Q26. Tungsten-carbide particles in a metal matrix (hard-facing/cermet) are used for:

(a) Light panels
(b) Extreme wear resistance on tool edges
(c) Wiring
(d) Cushioning

Answers

Q16 — (a) 2.7 g/cm³ (⅓ of steel)
Q17 — (b) Radiators, pistons, cylinder heads
Q18 — (b) Cu + Zn — *Brass = Cu + Zn (fittings).*
Q19 — (b) Cu + Sn — *Bronze = Cu + Sn.*
Q20 — (b) Good wear resistance and low friction — *Bronze for bearings/bushes — low friction.*
Q21 — (a) Zinc
Q22 — (b) Electrical/thermal conductor
Q23 — (b) A matrix plus a reinforcement
Q24 — (b) Sprayer/seed tanks, guards, hoods
Q25 — (b) High strength-to-weight and corrosion resistance
Q26 — (b) Extreme wear resistance on tool edges

Sources: Standard materials-science and manufacturing texts (Callister; Kalpakjian; Rajput — Engineering Materials); BIS/IS material specifications; SSAB/industry data on tillage-tool steels (as given in the source notes, July 2026); Kepner Ch. 3 & 7 (materials for tillage tools); NPTEL materials lectures.
